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MINI-REVIEW

A review of metabolic potential of human gut microbiome in human nutrition

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Abstract
The human gut contains a plethora of microbes, providing a platform for metabolic interaction between the host and microbiota. Metabolites produced by the gut microbiota act as a link between gut microbiota and its host. These metabolites act as messengers having the capacity to alter the gut microbiota. Recent advances in the characterization of the gut microbiota and its symbiotic relationship with the host have provided a platform to decode metabolic interactions. The human gut microbiota, a crucial component for dietary metabolism, is shaped by the genetic, epigenetic and dietary factors. The metabolic potential of gut microbiota explains its significance in host health and diseases. The knowledge of interactions between microbiota and host metabolism, as well as modification of microbial ecology, is really beneficial to have effective therapeutic treatments for many diet-related diseases in near future. This review cumulates the information to map the role of human gut microbiota in dietary component metabolism, the role of gut microbes derived metabolites in human health and host–microbe metabolic interactions in health and diseases.

Keywords Gut microbiota · Microbiome · Microbiota

Introduction

Humans are a permanent address for a number of microorganisms like bacteria, archaea, viruses, and unicellular eukaryotes, which altogether constitute microbiota, microflora or normal flora (Kunz et al. 2009; Neish 2009). Every part of the human body exposed to the external environment is inhabited by the microbiota (Neish 2009; Andersson et al. 2008). A number of efforts were made to elucidate the significance of the resident microbes, especially human gut microbes. The human gastrointestinal tract comprises of trillions of bacterial cells, approximately ten times more than the human cells (Ley et al. 2006). About 70% of the microbes identified from humans are found to be colonized in the gastrointestinal tract (Ley et al. 2006). The majority of them are Firmicutes, Bacteroidetes, and Proteobacteria, while Actinobacteria contribute less to the total bacterial composition. Bacteroidetes include the genus *Prevotella* and *Bacteroides*. Phylum Actinobacteria includes *Bifidobacterium* and phylum Firmicutes includes *Clostridium* clusters and members of *Eubacterium, Faecalibacterium, Roseburia*, and *Ruminococcus*. Humans have a critical relationship with the gut microbiota. Metagenomic studies have identified a huge and diverse catalog of microbial genes. The presence of these genes defines the role of gut microbiota in various metabolic pathways, complementing the host metabolic pathways. Molecular methods like metagenomics (Dinakaran et al. 2014; Monika et al. 2017) and metabolomics (Kordalewska and Markuszewski 2015) have made it easier to study the impact of the gut microbiota in various cardiovascular diseases, irritable bowel syndrome, diabetes, and cancer. A number of association studies link the human nutrition and health with the metabolic capacity of gut microbiota (Dutton and Turnbaugh 2012; Flint et al. 2012; Shanahan 2013). There occurs a commensalism between gut microbiota and host, where gut microbes use the substrate for fermentation fulfilling the nutrient and energy needs of the host (Louis et al. 2014a; Mondal et al. 2017). Bacterial cross-feeding is the mechanism that leads to the degradation of the complex and undigested food materials to energy-providing metabolites (Louis et al. 2014a). A symbiotic

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relationship is also seen in the case of rapid fucosylation of the host intestinal epithelium to sustain microbiota during sickness (Pickard et al. 2014). This symbiosis goes from simple to complex one, where microbial metabolites bind to various target proteins and alter the signaling pathways to possibly create metabolic and inflammation-related problems in humans (Kumar et al. 2017). The microbial community has diverse roles which include immune system maintenance, vitamin production, digestion, energy homeostasis, angiogenesis, metabolites synthesis and maintenance of intestinal barrier integrity (Dominguez-Bello and Blaser 2008; Rosenberg and Zilber-Rosenberg 2011). This review attempts to provide comprehensive information about human microbiome structure, function, and physiological roles.

Layout of human gut microbiota

Most of the human gut microbial communities comprise of the strict anaerobes outnumbering the facultative anaerobes and aerobes by a factor of two to three (Harris et al. 1976). Till now more than fifty bacterial phyla have been reported as human gut occupants (Schloss and Handelsman 2004), however, Bacteroidetes and Firmicutes are the dominant microbial groups (Turroni et al. 2008; Kumar et al. 2016), followed by Proteobacteria, Verrucomicrobia, Actinobacteria, Fusobacteria, and Cyanobacteria in lesser proportions (Eckburg et al. 2005). A qualitative and quantitative variation of these microbial groups was observed in respect to the studied region of the gastrointestinal tract (Fig. 1). The microbial communities of the throat and distal esophagus are majorly similar. Streptococcus, Prevotella, Actinomyces, Gemella, Rothia, Granulicatella, Haemophilus, and Veillonella are found to be the predominant microbes (Andersson et al. 2008). *Streptococcus* is the dominant genus at both the locations followed by the *Prevotella*. The remaining differentiating microbial community(< 25%) holds *Veillonella* (Andersson et al. 2008). Stomach microbial diversity depends upon presence and absence of *H. pylori* (Bik et al. 2006). Stomach lacking *H. pylori* contains *Streptococcus, Actinomyces, Prevotella*, and *Gemella*, predominantly found in the throat indicating that they may be transient residents swallowed from the throat (Andersson et al. 2008). Within *H. pylori* positive stomach, *H. pylori* dominates among the bacterial phyla, thus indicating the comfort adaptation in the stomach (Bik et al. 2006). The recto-sigmoid colon microbiota is far more complex than jejunum, ileum, and caecum resident microbes (Hayashi et al. 2005). *Enterococci, E. coli, Klebsiella, Lactobacilli, Staphylococci*, and *Streptococci* are present in the jejunum and ileum (Corrodi et al. 1978; Thadepalli et al. 1979). Most of the microbes of jejunum and ileum are aerobes and facultative anaerobes (Hayashi et al. 2005). Small intestine possesses aerobic *Enterococcus* group, *Lactobacilli, Streptococci*, and Gammaproteobacteria, while anaerobes are predominant in the large intestine (Hayashi et al. 2005). In contrast to this, *Bacteroides fragilis* and *Clostridium lituseburense* have been detected in the ileum (Wang et al. 2003). Caecal microbiota is more complex than jejunal and ileal microbiota. The main constituents of feces microbial community, *C. leptum, C. coccoides, and Bacteroides*, have also been found in the caecum (Hayashi et al. 2005). The ceacal microflora is predominated by *Lactobacillus, Enterococcus*, and *E. coli* (Marteau et al. 2001). Facultative aerobes form the predominant species of

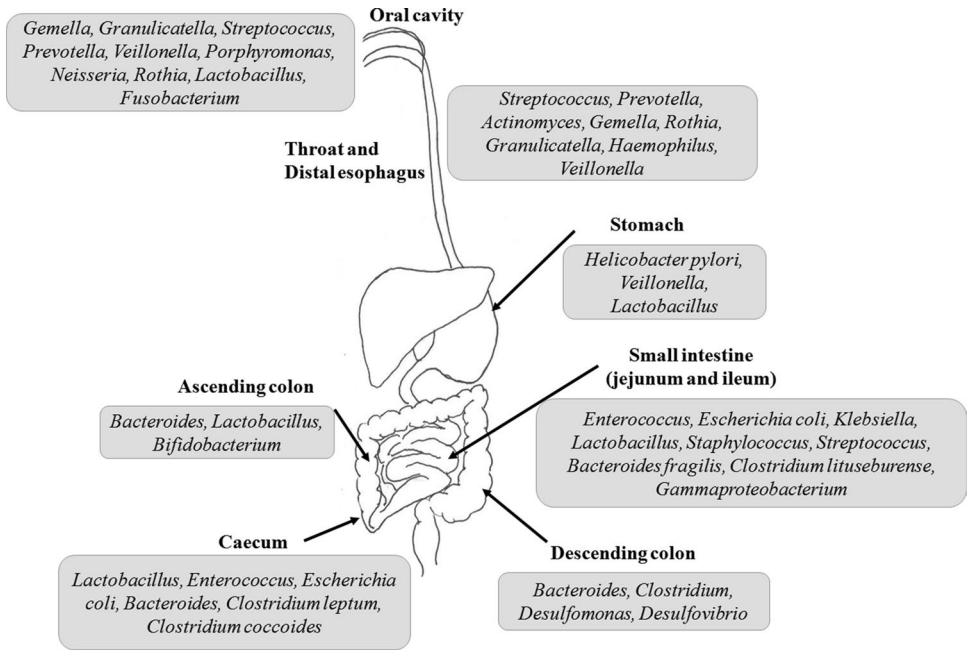


Fig. 1 Layout of human gut microbiome

the caecal microbial community (Hayashi et al. 2005). More complex microbial community, completely different from the caecal microbiota has been detected in the recto-sigmoidal colon (Hayashi et al. 2005). Strict anaerobic bacteria belonging to *Bacteroides, C. coccoides*, and *C. leptum* are predominant bacterial groups and subgroups (Hayashi et al. 2005). Predominant species vary among different individuals (Hayashi et al. 2002, 2003). Variation in the fecal microbiota among healthy and elder individuals also persists (Marteau et al. 2001; Hayashi et al. 2002). Gammaproteobacteria is highly frequent in the fecal microbiota of elders and healthy individuals (Hayashi et al. 2003). Species belonging to Gammaproteobacteria are also found in jejunum, ileum and caecum (Corrodi et al. 1978; Thadepalli et al. 1979) and a very rare number are present in the feces of healthy people (Hayashi et al. 2002).

Effect of diet on microbiota composition

Higher vertebrates have coevolved with their microbiota through an interaction between the gastrointestinal (GI) tract and microbial community. Therefore, any changes in the lifestyle, diet, and environmental conditions directly modulate the microbiota. The composition of the gut microbiota is also decided by the geographical factors and diet of the studied population. A study was designed to compare the microbiota of the children from a village Burkina Faso and the children from Italy (De Filippo et al. 2010). Till the breast-feeding stage, the microbiota of the two populations remained same; however, the introduction of solid food in the diet changed the scenario. The rural population showed an increase in Bacteroidetes responsible for degradation of polysaccharides, and decrease in the Firmicutes number due to attribution by the diet enriched with plant polysaccharides, while this feature was completely absent in the Italy resident children (De Filippo et al. 2010). The rural population showed the presence of bacteria from the genus *Xylanibacter* and *Prevotella*, which are responsible for xylan and cellulose hydrolysis. These bacteria were completely absent in urban residents (De Filippo et al. 2010). More SCFA (short-chain fatty acids) are produced in children on the fiber diet and when SCFA production varies with diet. These SCFA include acetate, propionate, and butyrate. Consequently, there was less amount of SCFA in Italy children than the African ones (De Filippo et al. 2010). The human genome has a direct relationship with the diet. People relying more on the starch-enriched diet were found to have more salivary amylase genes in their genome (Perry et al. 2007). A similar condition was seen in the Japanese people where they were found to have more porphyranase genes in the microbiota. Porphyranase genes encode for an enzyme that degrades porphyran, a carbohydrate majorly found in the seaweeds, a common food for Japanese people. Later on, various studies have proved that these genes were acquired by the gut microbiota from the microbes taken in along with the seaweeds by the process of horizontal gene transfer (Hehemann et al. 2010). A recent study describing the Indian gut microbial communities and comparing them with the microbiota from other populations shows the dominance of *Prevotella, Lactobacillus, Bifidobacterium*, and *Megasphaera* in Indian population (Bhute et al. 2016). Indian gut microbiota is found to variate from gut microbiota of Spanish, Korean, Bangladeshi and American populations due to different dietary habits and geographical locations (Bhute et al. 2016).

Infant diet and microbiota

Staphylococcus, Enterobacteriaceae, and *Streptococcus* are present in the infant feces. *Bifidobacterium* has been found at the third day of infant age (Sakata et al. 1985). As the aerobic bacteria in intestine gradually use oxygen thereby reducing the redox potential, the conditions become favorable for the anaerobic bacteria (Dominguez-Bello et al. 2011). Until 1 month of the infant age, the *Bifidobacterium* level in the intestine was somewhat similar in breast-fed and formula-fed infants (Penders et al. 2006). After the few months of age, microbial community dominating in the infant feeding breast-fed milk were of *Bifidobacteria* and lactic acid bacteria, while formula milk feeding infant had a complex microbiota including facultative anaerobes, *Bacteroides, Bifidobacteria* and *Clostridia* (Wall et al. 2009). Additionally, the number of pathogenic groups such as *Clostridium difficile, B. fragilis, E. coli* were more in the formula-fed infants than the breast-fed ones. This is because of the beneficial microbes attributed by the human milk nutrients. The mother milk contains oligosaccharides, a favored substrate of *Bifidobacteria*, thus increasing the number of *Bifidobacterial* species (Wall et al. 2009). It has been shown that breast-fed infants possess more protective bacteria from Actinobacteria class while formula-fed ones have more proinflammatory bacteria from Gammaproteobacteria class (Praveen et al. 2015). Introduction of solid foods in the infant's diet causes a drastic shift in the microbiota composition (Koenig et al. 2011).

Prenatal nutrition, stress, and microbiome

With the physical and mental maternal stress, a neuroendocrine immune response initiates (Sapolsky et al. 2000) that causes the release of proinflammatory cortisol and cytokines (Yang and Glaser 2002). Acute inflammation increases the risk of preterm birth (Ruma et al. 2007), while chronic

inflammation causes diabetes (Lowe et al. 2010), gestational hypertension (Freeman et al. 2004) and preterm birth (Giurgescu 2009). Maternal stress during pregnancy directly affects the neurodevelopment in the offspring (Weinstock 2005). Prenatal stress attributes to various adverse effects on the offspring, i.e., increased anxiety (Schneider et al. 2002), altered neurogenesis (Coe et al. 2002) and lack of attention in behavior (Keenan et al. 2013). Prenatal stress also has an effect on the hypothalamus–pituitary–adrenal axis of the offspring (Weinstock 1997). It has been found that the maternal stress during pregnancy is also associated with temperament related problems in offspring (Huizink et al. 2002), differences in the diurnal rhythms (Keenan et al. 2007), obstetric problems (Lou et al. 1994), shorter gestational period (Dayan et al. 2006) and shorter infant size at the time of birth (Wadhwa et al. 1993). Stress during pregnancy has an important effect on the infant's microbial composition (Zijlmans et al. 2015). Maternal stress and elevated cortisol concentration were found associated with an abundance of Proteobacterial groups related to pathogens such as *Escherichia, Serratia*, and *Enterobacter* along with lower relative abundance of *Bifidobacteria* and lactic acid bacteria (i.e., *Aerococcus Lactococcus, Lactobacillus*) in infant's intestine (Zijlmans et al. 2015). Additionally, a food supplementation study was performed in rural villages in Guatemala where two food supplements, fresco, and atole were given to pregnant, lactating mothers and children born from them up to 3 years of age. The prenatal nutrition has shown a direct effect on the children's early physical as well as mental development (Klein et al. 1976). At 36 months of the age, children who received more food supplements showed more growth retardation in comparison to the supplements deprived children (Klein et al. 1976). Microbial composition of the mother, genetic and other prenatal factors, as well as the mode of delivery influence the colonization of microbes in the infant's gut (Dominguez-Bello et al. 2010). Fetal microbiome starts developing in utero thus accounting for the important role of the maternal microbiome and other factors on infant's microbial composition (Kaplan et al. 2011). Infants delivered through vagina acquire vaginal microbes while those delivered through cesarean section acquire microbes present on the mother's skin (Penders et al. 2006; Dominguez-Bello et al. 2010). Along with this, preterm infants lack two of the important bacteria, *Bifidobacterium* and *Lactobacillus* and show the abundance of Proteobacteria (Barrett et al. 2013).

Metabolic potential of gut microbiota

Recently it has been proven that the gut metagenome has more coding capacity than the human genome. Coding capacity of the gut metagenome is about 150-fold higher than the human genome. Many of the biochemical pathways that humans do not have, are provided by the gut microbiota genome (Gill et al. 2006; Qin et al. 2010). Gut microbiota is associated with a majority of functions important for the host physiology and maintenance, i.e., development of the nervous system, intestinal development and regulation of appetite, etc. Gut microbiota is mainly associated with the degradation of the dietary fibers, proteins and peptides by fermentation and anaerobic degradation, respectively, to provide energy through metabolism (Acheson and Luccioli 2004), maintaining the gut health also (Xu and Gordon 2003). Formerly, the colon was known only to act as a storehouse for undigested food; however, the recent studies have proved the role of its inhabited microbes' in metabolism. The gut microbiota has been separately named as 'metabolic organ' because of its metabolic potential equivalent to the liver (O'Hara and Shanahan 2006). The main functions include energy-derivation from non-digestible carbohydrates, glycoconjugates derived from the host (glycosphingolipids), deconjugation and dehydroxylation of bile acids, biosynthesis of vitamins (K and B) and isoprenoids, cholesterol reduction and metabolism of amino acids and xenobiotics.

Gut microbes and carbohydrate metabolism

Humans do not have the capacity to degrade complex polysaccharides and non-digestible carbohydrates such as resistant starch, non-digestible polysaccharides and oligosaccharides and plant fibers, glycans obtained from the host (mucin) (Cummings and Englyst 1987). These undigested materials enter the colon and processed by the colon microbiota (Ouwehand et al. 2005). Normally, about 10–60 g of carbohydrate is taken in a diet. The majority of these carbohydrates are resistant starch (resistant to action by human amylase), non-starch polysaccharides, oligosaccharides and unabsorbed sugars such as raffinose, lactose and stachyose, etc. (Scott et al. 2008). The colonic microbes play an important role by producing different hydrolytic enzymes for the degradation of these substrates having complex structures of their respective macromolecules. A series of microbial processes are involved in the fermentation. Microbiota involved in these processes are characterized on the basis of the functional groups based on metabolite formed and substrate utilized. Though a skeptic information is available about the process and microbial partners in carbohydrates metabolism, still a metabolic sketch could be prepared. The complex carbohydrates are converted to polysaccharides by primary degradation, followed by their conversion to oligosaccharides (Rossi et al. 2005).

Initial degradation of the insoluble carbohydrates was performed by colonic microbes *Bacteroides* sp. and

Ruminococcus sp. (Flint et al. 2008). *Bifidobacterium* sp. and *Lactobacillus* sp. are responsible for the fermentation of oligosaccharides (fructooligosaccharides and glucooligosaccharides), polysaccharides (inulin). Major byproducts formed as a result of various microbial fermentation reactions are gases such as H_2, CO_2, methane and short-chain fatty acids (SCFA). Among various SCFA producers, members of Clostridium cluster XIVa are butyrate producers (Pryde et al. 2002). *Clostridium* cluster XIVa mediates butyrate production either by the CoA transferase or by comparatively less common pathway drawn by butyrate-kinase activity (Louis et al. 2004). Additionally, butyrate is also produced by Firmicutes such as *Faecalibacterium prausnitzii* and *Lachnospiraceae. Eubacterium hallii, Clostridium indolis* and *Anaerostipes caccae* also produce butyrate using L-lactate and D-lactate as the substrates (Duncan et al. 2004). Butyrate could be synthesized by many other bacterial groups, with no consistency within family genomes (Vital et al. 2014). Similarly, propionate could be synthesized by three routes, succinate pathway mediated by *Bacteroidetes* (Reichardt et al. 2014), propanediol pathway mediated by *Roseburia inulinivorans* (Mortensen and Clausen 1996; Reichardt et al. 2014) and succinate pathway (*Veillonella* spp.) and acrylate pathway (*Megasphaera elsdenii*) used by *Negativicutes* (Reichardt et al. 2014). Propionate and butyrate are also produced as a result of amino acid and peptide fermentation by Bacteroidetes and Firmicutes (Scott et al. 2006). Studies have shown that a particular bacterial group either produces propionate or butyrate and very few have the property to produce both (Reichardt et al. 2014). Saccharolytic fermentation results in generating beneficial metabolites, however, if there is the shortage of carbohydrates, the bacteria switch to alternate sources of energy usually resulting in products, harmful to humans (Boyd et al. 2013).

Carbohydrate fermentation is commonly associated with the evolution of gases namely, hydrogen, carbon dioxide, H_2S, and methane, (Louis et al. 2014b). Less than 1% of the gas is ammonia, skatole, indole, hydrogen sulfide and volatile amines. Oxygen makes only 1% of the total gas and total content of oxygen and nitrogen is only 26% (Suarez et al. 1997). Carbon dioxide is one of the major gases produced by the microbes and is routinely recycled in the body by acetogenesis and methanogenesis (Lajoie et al. 1988). It is not only produced by fermentation but also by other means such as bicarbonate acidification and the movement from the blood (Suarez et al. 1997). It is mainly produced by *Clostridium* species. The only source of hydrogen gas in the flatus is microbial fermentation (Wolf et al. 2016). It is mainly produced by *Clostridium* and *Bacteroides* (Christl et al. 1992). Higher energy can only be obtained by complete oxidation of the organic material, facilitated by the removal or recycling of hydrogen gas. Recycling of hydrogen takes place by three routes namely, acetogenesis, methanogenesis and sulfate reduction. Bacteria involved in sulfate reduction is *Desulfovibrio* (Gibson 1990), which uses sulfate as the substrate to remove hydrogen (Wolf et al. 2016). Methanogenesis and sulfate reduction both are the major routes to recycle hydrogen and the dominant route out of the two is decided by sulfate availability (Christl et al. 1993). Acetogenesis plays a role in hydrogen recycling acting as an additional route to either of the two above mentioned routes. However, acetogenesis is a beneficial way to recycle hydrogen and carbon dioxide as there is no emission of gas through this route (Christl et al. 1992). If any of the routes of gas recycling is hindered, it can lead to bloating due to increased gas production. A similar situation is seen in IBD and pneumatosis cystoides intestinalis (Macfarlane et al. 1986).

In a series of fermentation reactions, the metabolite of one microorganism acts as the substrate for the subsequent ones depending upon the energy produced or consumed (Samuel and Gordon 2006). It has been shown that metabolites as a result of degradation are not found in the feces, indicating their efficient utilization in the body. In healthy individuals, lactate produced is utilized by bacteria to from propionate or butyrate and thus lactate is absent in the feces; however, lactate has been found in the feces of patients with ulcerative colitis (Bjerrum et al. 2015), demarcating diseased condition. *Bifidobacterium longum* grown in the presence of fructo-oligosaccharide produces acetate, which is used by the *Roseburia intestinalis* and growth is stopped until it gets sufficient acetate during co-culture (Falony et al. 2006). Simultaneously, *B. longum* also produces lactate which is used by *Eubacterium hallii*, which is unable to grow alone on carbohydrate substrate. *Eubacterium hallii* produces butyrate as a result (Belenguer et al. 2006).

Gut microbes and protein metabolism

Protein turnover mainly occurs in the large intestine (Cummings and Macfarlane 1997) and colonic bacteria are highly efficient in degrading proteins from endogenous or exogenous sources (Gibson et al. 1989). *Bacteroides, Clostridium, Propionibacterium, Fusobacterium, Lactobacillus*, and *Streptococcus* were identified microbial species for their role in proteolysis (Macfarlane and Macfarlane 2006). The fecal microbiota has different proteolytic activity than the colonic microbiota (Smith and Macfarlane 1996). The activity of fecal microbes is high in terms of proteolysis of globular proteins only. A study with pure intestinal culture and intestinal culture inoculated with human feces demonstrated a reciprocal impact of fermentable carbohydrates onto amino acid fermentation (Russell et al. 2013). The protein fermentation was found elevated

with increased pH (Russell et al. 2013). Fermentation of aspartate, alanine, threonine, and methionine lead to the formation of propionate, while fermentation of glutamate, lysine, histidine, cysteine, serine, and methionine generates butyrate. Along with exogenous sources of proteins, mucus, exfoliated epithelial cells, as well as pancreatic enzymes also act as the protein source in the human colon. As a fermentation process, complex proteins are first cleaved by various bacterial peptidases, proteases, and endopeptidases to release free amino acids and short peptides (Macfarlane and Macfarlane 2006). Amino acids and short peptides then undergo fermentation. As a result of protein fermentation, branched chain fatty acids (2-methyl butyrate, isobutyrate, isovalerate), organic acids, gases (H_2 and CO_2) and trace amounts of phenols, amines, indoles, and ammonia are produced. The pathway for protein degradation has been recently described (Fig. 2) (Dai et al. 2010). Similar to the cross-feeding in case of carbohydrate fermentation, the concentration of one amino acid can affect the overall fermentation process (Hermanussen et al. 2010). Fermentation of arginine, aspartate and glycine are regulated by L-glutamine. Additionally, in the presence of L-glutamine catabolism of some non-essential and essential amino acids is reduced. Fermentation of proteins could generate toxins byproducts like amines, ammonia and branched chain fatty acids, thus protein fermentation renders the host with an increased risk of colon cancer (Bingham et al. 1996). One reason of association of colon cancer with protein degradation may be attributed to the fact that the most prevalent site in both the cases is the distal colon (Muir et al. 2004).

Gut microbes and vitamin synthesis

Bacteria inside the human gut synthesize a wide variety of vitamins, important for their own metabolism, as well as for the host to maintain host physiology. Studies in rodents and mammals have shown a role of microbes in the synthesis of vitamin B, vitamin K, pantothenic acid, riboflavin, thiamine, pyridoxine, nicotinic acid, folate, etc. (Gustafsson et al. 1962). The role of microbes in vitamin K production has been recently proven by comparing a rat living in normal condition with a rat living in sterile condition, fed on the diet lacking in vitamin K (Frick et al. 1967). Experiments on humans suggested that it is not the lack of vitamin K in the diet that affects prothrombin level, but it is an antibiotic treatment that diminishes the microbiota (LeBlanc et al. 2013). The metagenomic approach has greatly increased the knowledge about human gut microbes producing vitamins. A recent study demonstrated the enrichment of the microbiome with the Clustered Orthologous Groups (COGs) responsible for the formation of deoxyxylulose-5-phosphate, a precursor for certain vitamins (Magnúsdóttir et al. 2015). Analysis of human gut microbial genomes has identified the pathways for eight vitamins, possibly formed by the human microbiome. These eight vitamins from vitamin B group were niacin, biotin, riboflavin, thiamin, pantothenic acid, folate, pyridoxine, and cobalamine (Said 2013). Out of the commonly synthesized vitamins, riboflavin is the most prominent one, produced by 166 microbial groups, followed by niacin produced by another 162 groups. *Bacteroidetes* was

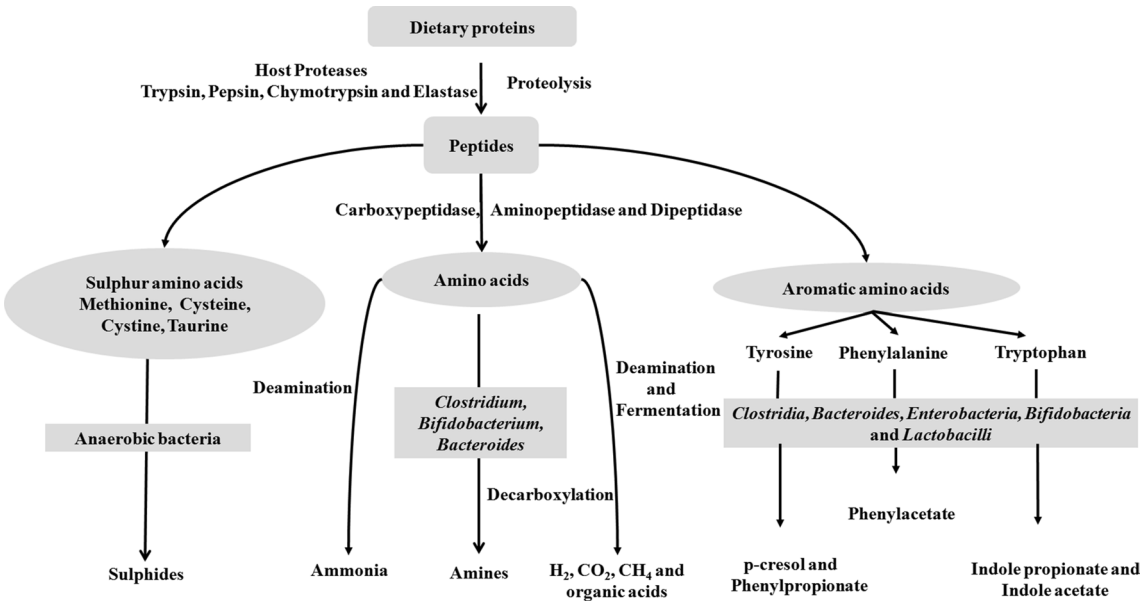


Fig. 2 Scheme diagram representing the role of human gut microbes in dietary protein catabolism

the major group involved in the synthesis of vitamin B. Riboflavin and biotin were synthesized mostly by *Proteobacteria, Fusobacteria* and *Bacteroidetes*, followed by *Actinobacteria* and *Firmicutes*. Complementation was also found during vitamin synthesis (Said 2013), indicating cross-feeding among bacteria. Gut bacteria provide a pool of vitamins in addition to the dietary source (Said 2013).

Gut microbes and polyphenol metabolism

About 820 mg/day of the polyphenol is ingested through diet (Pérez-Jiménez et al. 2011). There are various types of polyphenols in terms of their presence, metabolism, and roles (Manach et al. 2004). Major categories are flavonoids, phenolic acids, stilbenes, secoiridoids and lignans (Marín et al. 2015). Majority of these polyphenols enter the colon without being absorbed in the small intestine (Manach et al. 2004). The polyphenols are degraded to their metabolites by the colonic microbiota (Duda-Chodak et al. 2015). *Bacteroides distasonis, Bacteroides ovatus, Bacteroides uniformis, Enterococcus casseliflavus, Eubacterium cellulosolvens, Eubacterium ramulus* and *Lachnospiraceae CG191* are the species involved in primary degradation (hydrolysis) of the polyphenols (Marín et al. 2015; Braune et al. 2016). After initial hydrolysis, the resultant monomers and the aglycones were further metabolized via decarboxylation and ring-cleavage to form simpler forms such as hydroxyphenyl propionic acid and hydroxyphenyl acetic acids. These reactions have been seen for flavonoids, lignans and phenolic acids (Braune and Blaut 2016; Braune et al. 2016). All the reactions of polyphenol degradation are mediated by a large number of bacterial groups in a stepwise manner. In the case of secoisolariciresinol diglucoside degradation, *B. distasonis, B. fragilis, B. ovatus, C. cocleatum* and *C. saccharogumia* are involved in the primary step (deglycosylation). Resultant aglycone is demethylated by *Blautia, Butyribacterium methylotrophicum, Eubacterium callanderi, Eubacterium limosu*, and *Peptostreptococcus* productus. Further dehydroxylation is carried out by *Clostridium scindens* and *Eggerthella lenta*, followed by dehydrogenation and ring closure by *Clostridiales*, in particular, *Lactonifactor longoviformis* (Quartieri et al. 2016). Inter-individual variation has been seen in the various polyphenol metabolism, absorption, and excretion (Tomas-Barberan et al. 2014; Quartieri et al. 2016). Such variation has been seen in the case of daidzein, an isoflavone. Daidzein can be metabolized by two pathways (Rechner et al. 2004). In one pathway, *Clostridium* converts daidzein to O-desmethylangolensin (major pathway) and in another pathway, daidzein is metabolized by a consortium of microbes like *Streptococcus intermedius, B. ovatus, Rumino coccus productus, Eggerthella* sp. *Julong 732, Adlercreutzia equolifaciens, Slakia isoflavoniconvertens*, and *Slakia equolifaciens* to convert daidzein to (S)-equol via dihydrodaidzein and tetrahydrodaidzein. *Enterococcus faecium, Lactobacillus mucosae*, and *Finegoldia magna EPI3, Veillonella* sp. can have an effect on this pathway (Decroos et al. 2005). The ellagic acids and ellagitannins are converted to urothilin conjugates or isourothilin or none (Tomas-Barberan et al. 2014). This variation can have beneficial effects on the host. In the case of daidzein degradation, equol is more bioactive than daidzein itself and has anti-oxidant, anti-cancerous as well as oestrogenic and anti-oestrogenic effects (Landete et al. 2016). Equol has been shown to have effects on bone health, menopause and breast cancer (Lampe 2009). Likewise, urothilin is better absorbed and is beneficial than its parent ellagitannin (Tomás-Barberán et al. 2017).

Gut microbes and bile acid metabolism

Synthesis of bile acids takes place in the liver and help in lipid digestion. Bile synthesis in liver occurs from chenodeoxycholic acid and cholic acid. The chenodeoxycholic acid is converted to lithocholic acid and cholic acid is converted to deoxycholic acid (Gustafsson et al. 1977). Bile acids are made up of steroid nucleus with one or more side chains. The microbes metabolize both the side chain, as well as the steroid nucleus. Gut bacteria contain bile salt hydrolase enzyme to break the amide bond between a bile acid and its conjugate amino acid (Kellogg 1971). *Bacteroides, Bifidobacterium, Clostridium, Lactobacillus*, and *Listeria* are known to have this hydrolase enzyme-coding genes (Jones et al. 2008). In this way, bacteria detoxify the bile acids (Ridlon et al. 2006). This deconjugated form of bile acids is then recycled by absorption in the liver. Microbial degradation of the steroid nucleus results in the dehydroxylation mostly at the C_7 position. Microbes also modify the bile acids by epimerization of hydroxyl group. The 7α hydroxyl group of chenodeoxycholic acid is epimerized to form non-toxic ursodeoxycholic acid (Woollett et al. 2003). On an average, thirty diverse bile acids circulate within the human body. This diversity is maintained by the gut microbiota (García-Cañaveras et al. 2012).

Gut microbes and choline metabolism

Choline is mainly ingested through red meat and eggs. It is also synthesized in the human body. It is important for the synthesis of very low-density lipoproteins and lipid metabolism. Lack of choline in the diet has a direct effect on gut microbes and can cause liver steatosis in mice as well as humans (Dumas et al. 2006; Spencer et al. 2011). Hepatic steatosis leads to lesser Gammaproteobacteria and more *Erysipelotrichi* in the fecal microbiota

(Dumas et al. 2006). Microbes, as well as host enzymes, are responsible for choline degradation. Microbiota converts choline to trimethylamine, which is then converted to trimethyl-*N*-oxide by the hepatic enzymes (Prentiss et al. 1961; Wang et al. 2011). This can lead to non-alcoholic fatty-acid liver disease in mice (Prentiss et al. 1961). The presence of trimethyl-*N*-oxide and its metabolism is associated with cardiovascular heart diseases (Wang et al. 2011).

Gut microbes and aromatic amino acid metabolism

The intestinal microbiota is also associated with aromatic amino acid metabolism and plays a fundamental role in the production of different types of phenolic and indolic metabolites by transamination, decarboxylation, deamination, and dehydrogenation. Aromatic amino acids are fermented by bacteria like *Clostridium, Bacteroides, Bifidobacterium, Lactobacillus*, and *Peptostreptococcus*. In the human body, large amounts of indole are generated and converted to indoxyl-sulfate, which is excreted through urine. The indole is generated by tryptophanase (TnA) which reversibly converts tryptophan into indole, ammonia, and pyruvate. In the human gut, more than 85 bacterial species are known as indole producing bacteria. The indoles are used to strengthen host cell-barrier properties, increase resistance to pathogen colonization and maintain controlled inflammation (Wauson et al. 2013). Aromatic amino acids (tyrosine, phenylacetic acid, and tryptophan) are degraded to phenylacetic acid and 4-hydroxyphenyl acetic acid (Zelante et al. 2013). Tryptophan, an essential amino acid, is metabolized to indole by the gut-residing bacteria using the enzyme tryptophanase (Kim and Park 2013). Tryptophan is converted to indole-3-aldehyde by *Lactobacillus* spp. through undefined enzymes (Zelante et al. 2013). *Clostridium sporogenes* uses tryptophan deaminase to convert tryptophan to Indole propionic acid (IPA) (Wikoff et al. 2009). Almost 90% of the tryptophan is catabolized to kynurenine (Dougherty et al. 2008). Kynurenine results in the formation of kynurenic acid and quinolinic acid. Quinolinic acid and kynurenic acid act as agonist and antagonist of glutamate receptor, respectively (Moroni 1999). Approximately, 3% of the total tryptophan is used for the synthesis of serotonin in the body (Carroll 1971), out of which only 1% is used in the brain for serotonin synthesis (Sandyk 1992). Tryptamine is formed from the decarboxylation of tryptophan. Tryptamine acts as the regulator of serotonin functions (Jones 1982).

Sulfur metabolism in the gut

Many aspects of human health and cellular functions are associated with sulfur amino acids, cysteine, and methionine metabolism. There are two ways for the production of hydrogen sulfide in the body. One is amino acid fermentation and the second is bacterial sulfate reduction. Sulfur-containing amino acids like methionine, cystine, cysteine, taurine and inorganic sulfate as well as sulfated mucin are main sources of the sulfur in the diet. Other dietary sources are vegetables like broccoli, kales, and cabbages. *Desulfovibrio* is the main sulfur-reducing bacteria (SRB) (Loubinoux et al. 2002; Scanlan et al. 2009), which is present in the human gut and it utilizes the sulfate or sulfite ion as an electron acceptor intended for the reduced organic substrates dissimulation (Cord-Ruwisch et al. 1988). Sulfide is produced in the small intestine lumen and hydrolyzed to H_2S (Cord-Ruwisch et al. 1988). There are till now 60 genera and 220 species known to reduce sulfate (Muyzer and Stams 2008; Leloup et al. 2009).

The role of gut microbes in host physiology

Along with facilitating the metabolic process in the human body, gut microbiota is required for a number of physiological effect on its host, such as immune system development, to confer protection against pathogens as well as for host's intestinal epithelium development and maintaining tissue homeostasis. Microbes also help in the development of bones (Sjögren et al. 2012). The properties of the mucus layer depend on the microbial composition (Juge 2012). Although the bacteria are found in the outer mucus layer (Johansson et al. 2008), however, the entire mucus layer shows the microbial population which attaches to the layer via glycosidases or lectins (Derrien et al. 2010; Juge 2012). Gut microbiota influence the clustering of lymphoid tissue inducer (LTi) called cryopatches (Kanamori et al. 1996), which recruit B cells and form lymphoid follicles required for intestinal immunity (Eberl 2005). Gut microbiota help in the immune cell differentiation by systematically shaping the T-cell landscape and thus maintaining homeostasis (Kieper et al. 2005). Along with this, gut microbiota regulates cytokine and chemokine production, as well as maintains other immune mediators (Uematsu et al. 2008). Gut microbiota also regulates AMP production (Vaishnava et al. 2011). Intestinal microbiota has a profound effect on the development of gastrointestinal tract (Reinhardt et al. 2012). Remodeling of the vascular system is also among one of the attributions of the microbiota to its host (Stappenbeck et al. 2002). Microorganisms modulate the

Table 1 Various studies indicating potential role of human gut microbiome in host physiology

Microbial benefits to humans	References
Development of nervous system	(Principi and Esposito 2016)
Maturation and education of immune system	(Chung et al. 2012; Kamada and Nunez 2013)
Protection against pathogens	(Kamada et al. 2013; Ubeda et al. 2017)
Renewal of intestinal epithelial layer	(Koboziev et al. 2014; Lin and Zhang 2017)
Regulation of intestinal barrier integrity	(Sharma et al. 2010)
Regulation of appetite and behavior	(Norris et al. 2013; Kanwar et al. 2016; Fetissov 2017)
Intestinal angiogenesis	(Stappenbeck et al. 2002)
Recovery of intestinal epithelial injury	(Yu et al. 2012)
Improved energy harvest through digestion of complex fibers in food	(Savage 1970; Hooper et al. 1998; Guarner and Malagelada 2003; Xu and Gordon 2003; Acheson and Luccioli 2004)
Production of vitamins	(Frick et al. 1967b; Gustafsson et al. 1977; Hill 1997; LeBlanc et al. 2013; Magnúsdóttir et al. 2015)
Metabolism of xenobiotics	(O'Hara and Shanahan 2006)

Table 2 Association of human gut microbes in the onset of various human disease

Disease	Dysbiosis	References
Psoriasis	Increased ratio of *Firmicutes* to *Actinobacteria*	(Gao et al. 2008)
Reflux oesophagitis	Oesophageal microbiota dominated by gram-negative anaerobes; gastric microbiota with low or absent *Helicobacter pylori*	(Peek and Blaser 2002; Islami and Kamangar 2008)
Obesity	Reduced ratio of *Bacteroidetes* to *Firmicutes*	(Ley et al. 2005; Turnbaugh et al. 2006)
Childhood-onset asthma	Absent gastric *H. pylori* (especially the cytotoxin-associated gene A (cagA) genotype)	(Chen and Blaser 2007)
Inflammatory bowel disease (colitis), Ulcerative colitis and Crohn's disease	Larger populations of *Enterobacteriaceae* *R. hominis* and *F. prausnitzii* decrease in number *Bacteroides* and *Bifidobacteria* decrease in number	(Garrett et al. 2010; Machiels et al. 2014; Seksik et al. 2003)
Functional bowel diseases	Larger populations of *Veillonella* and *Lactobacillus*	(Tana et al. 2010)
Colorectal carcinoma	Larger populations of *Fusobacterium* spp., *Acidaminobacter, Phascolarctobacterium, Citrobacter farmer, Akkermansia muciniphila* and low number of *Prevotella, Ruminococcus* spp	(Kostic et al. 2012; Weir et al. 2013)
Cardiovascular disease	Gut microbiota-dependent metabolism of phosphatidylcholine. *Desulfovibrio desulfuricans* produce TMA. Phyla *Firmicutes* and *Proteobacteria* and six genera that showed significant choline consumption and TMA accumulation: *Anaerococcus hydrogenalis, Clostridium asparagiforme, Clostridium sporogenes, Escherichia fergusonii, Clostridium hathewayi, Proteus penneri, Providencia rettgeri*, and *Edwardsiella tarda*	(Wang et al. 2010; Koeth et al. 2013; Romano et al. 2015)
Type-I diabetes	Reduced number of *Lactobacillus, Bifidobacterium, Blautia coccoides, Eubacterium rectal, Prevotella* and increased number of *Clostridium, Bacteroides, Veillonella*	(Murri et al. 2013; Zhang et al. 2015)
Type-II diabetes	Reduced *Clostridia, Firmicutes*. Ratio of *Bacteroides* to *Firmicutes* are significantly positive	
Rheumatic arthritis	Low number of *Bifidobacteria* and *Bacteroides fragilis*	(Yeoh et al. 2013)
Autism (ASDs)	Excessive growth of *Clostridia* and decrease in *Bifidobacteria*	(Heberling et al. 2013)
Liver disease (NASH)	Excessively increased *E.coli*	(Chu et al. 2004)
COPD	*Proteobacteria* in COPD and asthmatic airways *Firmicute* in COPD was increased, while diminution in *Bacteriodetes* in COPD patients	

inflammation thereby influencing carcinogenesis (Hope et al. 2005) (Table 1).

Human gut microbiome associated with diseases

Many different types of microbial communities are associated with a specific disease or even temporary changes in microbiota are correlated during course of a disease. In different conditions, it is quite challenging to discover a causal link between the differential abundance of microbes and pathological symptoms. Commensalism is also present in human microbes and the host. Other than this, human gut bacteria are also associated with different diseases due to abnormal changes in the gut ecosystem (Zhang et al. 2015) (Table 2).

Conclusion

Gut microbiota is an organ responsible for touching almost every functional aspect of its host. Diet has a major role in affecting the microbial composition. The alterations in the diet may result in serious pathological conditions as well. However, alterations in the microbiota composition can simultaneously be used in treating the metabolic diseases. Here, we have demonstrated the gut microbiota composition and its role in the metabolism of the dietary nutrients. We have also shown the effects of the microbiota on the host physiology. Carbohydrate fermentation is an important metabolic pathway in humans and the metabolites produced have a major effect on the host. Along with this, the pathways for metabolism of other macromolecules have also been shown. Microbiota also has the property to degrade the polyphenols and different individuals have different effects of the polyphenols because of differential metabolism due to variation in microbiota composition among individuals. Alterations in the gut microbiota composition can be harmful such as the decrease in lactate and butyrate-producing bacteria lead to decrease in mucin synthesis and is found responsible for autoimmune diseases leading to diabetes.

Possible future perspectives

Dissimilarity and variations in human microbiota in several body habitats and their metabolic interactions with host will stimulate key insight into the role of gut microbiome for healthy host physiology. Differential analysis of healthy and diseased host gut microbiome at metagenome, metatranscriptome and metabolome level could be used for prediction and management of systemic diseases. Eventually, this information will lead to the identification of next-generation diagnostics and therapeutics.

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Compliance with ethical standards

Conflict of interest No conflict of interest exists.

References

Acheson DWK, Luccioli S (2004) Microbial–gut interactions in health and disease. Mucosal immune responses. Best Pract Res Clin Gastroenterol 18:387–404. doi: https://doi.org/10.1016/j.bpg.2003.11.002

Andersson AF, Lindberg M, Jakobsson H et al (2008) Comparative analysis of human gut microbiota by barcoded pyrosequencing. PloS One 3:e2836. doi: https://doi.org/10.1371/journal.pone.0002836

Barrett E, Kerr C, Murphy K et al (2013) The individual-specific and diverse nature of the preterm infant microbiota. Arch Dis Child Fetal Neonatal Ed 98:F334–F340. doi: https://doi.org/10.1136/archdischild-2012-303035

Belenguer A, Duncan SH, Calder AG et al (2006) Two routes of metabolic cross-feeding between *Bifidobacterium* adolescentis and butyrate-producing anaerobes from the human gut. Appl Environ Microbiol 72:3593–3599. doi: https://doi.org/10.1128/AEM.72.5.3593-3599.2006

Bhute S, Pande P, Shetty SA et al (2016) Molecular characterization and meta-analysis of gut microbial communities illustrate enrichment of *Prevotella* and *Megasphaera* in Indian subjects. Front Microbiol. doi: https://doi.org/10.3389/fmicb.2016.00660

Bik EM, Eckburg PB, Gill SR et al (2006) Molecular analysis of the bacterial microbiota in the human stomach. Proc Natl Acad Sci USA 103:732–737. doi: https://doi.org/10.1073/pnas.0506655103

Bingham SA, Pignatelli B, Pollock JR et al (1996) Does increased endogenous formation of *N*-nitroso compounds in the human colon explain the association between red meat and colon cancer? Carcinogenesis 17:515–523

Bjerrum JT, Wang Y, Hao F et al (2015) Metabonomics of human fecal extracts characterize ulcerative colitis, Crohn's disease and healthy individuals. Metab Off J Metab Soc 11:122–133. doi: https://doi.org/10.1007/s11306-014-0677-3

Boyd SD, Liu Y, Wang C et al (2013) Human lymphocyte repertoires in ageing. Curr Opin Immunol 25:511–515. doi: https://doi.org/10.1016/j.coi.2013.07.007

Braune A, Blaut M (2016) Bacterial species involved in the conversion of dietary flavonoids in the human gut. Gut Microbes 7:216–234. doi: https://doi.org/10.1080/19490976.2016.1158395

Braune A, Engst W, Blaut M (2016) Identification and functional expression of genes encoding flavonoid O- and C-glycosidases in intestinal bacteria. Environ Microbiol 18:2117–2129. doi: https://doi.org/10.1111/1462-2920.12864

Carroll BJ (1971) Monoamine precursors in the treatment of depression. Clin Pharmacol Ther 12:743–761. doi: https://doi.org/10.1002/cpt1971125743

Chen Y, Blaser MJ (2007) Inverse associations of *Helicobacter pylori* with asthma and allergy. Arch Intern Med 167:821–827. doi: https://doi.org/10.1001/archinte.167.8.821

Christl SU, Murgatroyd PR, Gibson GR, Cummings JH (1992) Production, metabolism, and excretion of hydrogen in the large intestine. Gastroenterology 102:1269–1277

Christl SU, Gibson GR, Murgatroyd PR et al (1993) Impaired hydrogen metabolism in pneumatosis cystoides intestinalis. Gastroenterology 104:392–397

Chu F-F, Esworthy RS, Chu PG et al (2004) Bacteria-induced intestinal cancer in mice with disrupted Gpx1 and Gpx2 genes. Cancer Res 64:962–968

Chung H, Pamp SJ, Hill JA et al (2012) Gut immune maturation depends on colonization with a host-specific microbiota. Cell 149(7):1578–1593

Coe CL, Lulbach GR, Schneider ML (2002) Prenatal disturbance alters the size of the corpus callosum in young monkeys. Dev Psychobiol 41:178–185. doi: https://doi.org/10.1002/dev.10063

Cord-Ruwisch R, Seitz H-J, Conrad R (1988) The capacity of hydrogenotrophic anaerobic bacteria to compete for traces of hydrogen depends on the redox potential of the terminal electron acceptor. Arch Microbiol 149:350–357. doi: https://doi.org/10.1007/BF00411655

Corrodi P, Wideman PA, Sutter VL et al (1978) Bacterial flora of the small bowel before and after bypass procedure for morbid obesity. J Infect Dis 137:1–6

Cummings JH, Englyst HN (1987) Fermentation in the human large intestine and the available substrates. Am J Clin Nutr 45:1243–1255

Cummings JH, Macfarlane GT (1997) Role of intestinal bacteria in nutrient metabolism. Clin Nutr 16:3–11. doi: https://doi.org/10.1016/S0261-5614(97)80252-X

Dai Z-L, Zhang J, Wu G, Zhu W-Y (2010) Utilization of amino acids by bacteria from the pig small intestine. Amino Acids 39:1201–1215. doi: https://doi.org/10.1007/s00726-010-0556-9

Dayan J, Creveuil C, Marks MN et al (2006) Prenatal depression, prenatal anxiety, and spontaneous preterm birth: a prospective cohort study among women with early and regular care. Psychosom Med 68:938–946. doi: https://doi.org/10.1097/01.psy.0000244025.20549.bd

De Filippo C, Cavalieri D, Di Paola M et al (2010) Impact of diet in shaping gut microbiota revealed by a comparative study in children from Europe and rural Africa. Proc Natl Acad Sci USA 107:14691–14696. doi: https://doi.org/10.1073/pnas.1005963107

Decroos K, Vanhemmens S, Cattoir S et al (2005) Isolation and characterisation of an equol-producing mixed microbial culture from a human faecal sample and its activity under gastrointestinal conditions. Arch Microbiol 183:45–55. doi: https://doi.org/10.1007/s00203-004-0747-4

Derrien M, van Passel MW, van de Bovenkamp JH et al (2010) Mucin-bacterial interactions in the human oral cavity and digestive tract. Gut Microbes 1:254–268. doi: https://doi.org/10.4161/gmic.1.4.12778

Dinakaran V, Rathinavel A, Pushpanathan M et al (2014) Elevated levels of circulating DNA in cardiovascular disease patients: metagenomic profiling of microbiome in the circulation. PLoS One 9:e105221. doi: https://doi.org/10.1371/journal.pone.0105221

Dominguez-Bello MG, Blaser MJ (2008) Do you have a probiotic in your future? Microbes Infect 10:1072–1076. doi: https://doi.org/10.1016/j.micinf.2008.07.036

Dominguez-Bello MG, Costello EK, Contreras M et al (2010) Delivery mode shapes the acquisition and structure of the initial microbiota across multiple body habitats in newborns. Proc Natl Acad Sci USA 107:11971–11975. doi: https://doi.org/10.1073/pnas.1002601107

Dominguez-Bello MG, Blaser MJ, Ley RE, Knight R (2011) Development of the human gastrointestinal microbiota and insights from high-throughput sequencing. Gastroenterology 140:1713–1719. doi: https://doi.org/10.1053/j.gastro.2011.02.011

Dougherty DM, Marsh-Richard DM, Mathias CW et al (2008) Comparison of 50- and 100-g L-tryptophan depletion and loading formulations for altering 5-HT synthesis: pharmacokinetics, side effects, and mood states. Psychopharmacology 198:431–445. doi: https://doi.org/10.1007/s00213-008-1163-2

Duda-Chodak A, Tarko T, Satora P, Sroka P (2015) Interaction of dietary compounds, especially polyphenols, with the intestinal microbiota: a review. Eur J Nutr 54:325–341. doi: https://doi.org/10.1007/s00394-015-0852-y

Dumas M-E, Barton RH, Toye A et al (2006) Metabolic profiling reveals a contribution of gut microbiota to fatty liver phenotype in insulin-resistant mice. Proc Natl Acad Sci USA 103:12511–12516. https://doi.org/10.1073/pnas.0601056103

Duncan SH, Louis P, Flint HJ (2004) Lactate-utilizing bacteria, isolated from human feces, that produce butyrate as a major fermentation product. Appl Environ Microbiol 70:5810–5817. doi: https://doi.org/10.1128/AEM.70.10.5810-5817.2004

Dutton RJ, Turnbaugh PJ (2012) Taking a metagenomic view of human nutrition. Curr Opin Clin Nutr Metab Care 15:448–454. doi: https://doi.org/10.1097/MCO.0b013e3283561133

Eberl G (2005) Inducible lymphoid tissues in the adult gut: recapitulation of a fetal developmental pathway? Nat Rev Immunol 5:413–420. doi: https://doi.org/10.1038/nri1600

Eckburg PB, Bik EM, Bernstein CN et al (2005) Diversity of the human intestinal microbial flora. Science 308:1635–1638. doi: https://doi.org/10.1126/science.1110591

Falony G, Vlachou A, Verbrugghe K, De Vuyst L (2006) Cross-feeding between Bifidobacterium longum BB536 and acetate-converting, butyrate-producing colon bacteria during growth on oligofructose. Appl Environ Microbiol 72:7835–7841. doi: https://doi.org/10.1128/AEM.01296-06

Fetissov SO (2017) Role of the gut microbiota in host appetite control: bacterial growth to animal feeding behaviour. Nat Rev Endocrinol 13(1):11–25

Flint HJ, Bayer EA, Rincon MT et al (2008) Polysaccharide utilization by gut bacteria: potential for new insights from genomic analysis. Nat Rev Microbiol 6:121–131. doi: https://doi.org/10.1038/nrmicro1817

Flint HJ, Scott KP, Louis P, Duncan SH (2012) The role of the gut microbiota in nutrition and health. Nat Rev Gastroenterol Hepatol 9:577–589. doi: https://doi.org/10.1038/nrgastro.2012.156

Freeman DJ, McManus F, Brown EA et al (2004) Short- and long-term changes in plasma inflammatory markers associated with preeclampsia. Hypertens Dallas Tex 1979 44:708–714. doi: https://doi.org/10.1161/01.HYP.0000143849.67254.ca

Frick PG, Riedler G, Brögli H (1967) Dose response and minimal daily requirement for vitamin K in man. J Appl Physiol 23:387–389

Gao Z, Tseng C, Strober BE et al (2008) Substantial alterations of the cutaneous bacterial biota in psoriatic lesions. PloS One 3:e2719. doi: https://doi.org/10.1371/journal.pone.0002719

García-Cañaveras JC, Donato MT, Castell JV, Lahoz A (2012) Targeted profiling of circulating and hepatic bile acids in human, mouse, and rat using a UPLC-MRM-MS-validated method. J Lipid Res 53:2231–2241. doi: https://doi.org/10.1194/jlr.D028803

Garrett WS, Gallini CA, Yatsunenko T et al (2010) Enterobacteriaceae act in concert with the gut microbiota to induce spontaneous and maternally transmitted colitis. Cell Host Microbe 8:292–300. doi: https://doi.org/10.1016/j.chom.2010.08.004

Gibson GR (1990) Physiology and ecology of the sulphate-reducing bacteria. J Appl Bacteriol 69:769–797

Gibson SA, McFarlan C, Hay S, MacFarlane GT (1989) Significance of microflora in proteolysis in the colon. Appl Environ Microbiol 55:679–683

Gill SR, Pop M, Deboy RT et al (2006) Metagenomic analysis of the human distal gut microbiome. Science 312:1355–1359. doi: https://doi.org/10.1126/science.1124234

Giurgescu C (2009) Are maternal cortisol levels related to preterm birth? J Obstet Gynecol Neonatal Nurs JOGNN 38:377–390. doi: https://doi.org/10.1111/j.1552-6909.2009.01034.x

Guarner F, Malagelada J-R (2003) Gut flora in health and disease. Lancet 361(9356):512–519

Gustafsson BE, Daft FS, McDANIEL EG et al (1962) Effects of vitamin K-active compounds and intestinal microorganisms in vitamin K-deficient germfree rats. J Nutr 78:461–468

Gustafsson BE, Angelin B, Einarsson K, Gustafsson JA (1977) Effects of cholesterol feeding on synthesis and metabolism of cholesterol and bile acids in germfree rats. J Lipid Res 18:717–721

Harris MA, Reddy CA, Carter GR (1976) Anaerobic bacteria from the large intestine of mice. Appl Environ Microbiol 31:907–912

Hayashi H, Sakamoto M, Benno Y (2002) Phylogenetic analysis of the human gut microbiota using 16S rDNA clone libraries and strictly anaerobic culture-based methods. Microbiol Immunol 46:535–548

Hayashi H, Sakamoto M, Kitahara M, Benno Y (2003) Molecular analysis of fecal microbiota in elderly individuals using 16S rDNA library and T-RFLP. Microbiol Immunol 47:557–570

Hayashi H, Takahashi R, Nishi T et al (2005) Molecular analysis of jejunal, ileal, caecal and recto-sigmoidal human colonic microbiota using 16S rRNA gene libraries and terminal restriction fragment length polymorphism. J Med Microbiol 54:1093–1101. doi: https://doi.org/10.1099/jmm.0.45935-0

Heberling CA, Dhurjati PS, Sasser M (2013) Hypothesis for a systems connectivity model of Autism Spectrum Disorder pathogenesis: links to gut bacteria, oxidative stress, and intestinal permeability. Med Hypotheses 80:264–270. doi: https://doi.org/10.1016/j.mehy.2012.11.044

Hehemann J-H, Correc G, Barbeyron T et al (2010) Transfer of carbohydrate-active enzymes from marine bacteria to Japanese gut microbiota. Nature 464:908–912. doi: https://doi.org/10.1038/nature08937

Hermanussen M, Gonder U, Jakobs C et al (2010) Patterns of free amino acids in German convenience food products: marked mismatch between label information and composition. Eur J Clin Nutr 64:88–98. doi: https://doi.org/10.1038/ejcn.2009.116

Hill MJ (1997) Intestinal flora and endogenous vitamin synthesis. Eur J Cancer Prev 6(Suppl 1):S43–S45

Hooper LV, Bry L, Falk PG, Gordon JI (1998) Host-microbial symbiosis in the mammalian intestine: exploring an internal ecosystem. BioEssays 20(4):336–343. doi: https://doi.org/10.1002/(SICI)1521-1878(199804)20:4<336::AID-BIES10>3.0.CO;2-3

Hope ME, Hold GL, Kain R, El-Omar EM (2005) Sporadic colorectal cancer—role of the commensal microbiota. FEMS Microbiol Lett 244:1–7. doi: https://doi.org/10.1016/j.femsle.2005.01.029

Huizink AC, de Medina PGR, Mulder EJH et al (2002) Psychological measures of prenatal stress as predictors of infant temperament. J Am Acad Child Adolesc Psychiatry 41:1078–1085

Islami F, Kamangar F (2008) *Helicobacter pylori* and esophageal cancer risk: a meta-analysis. Cancer Prev Res Phila Pa 1:329–338. doi: https://doi.org/10.1158/1940-6207.CAPR-08-0109

Johansson MEV, Phillipson M, Petersson J et al (2008) The inner of the two Muc2 mucin-dependent mucus layers in colon is devoid of bacteria. Proc Natl Acad Sci USA 105:15064–15069. doi: https://doi.org/10.1073/pnas.0803124105

Jones RS (1982) Tryptamine: a neuromodulator or neurotransmitter in mammalian brain? Prog Neurobiol 19:117–139

Jones BV, Begley M, Hill C et al (2008) Functional and comparative metagenomic analysis of bile salt hydrolase activity in the human gut microbiome. Proc Natl Acad Sci USA 105:13580–13585. doi: https://doi.org/10.1073/pnas.0804437105

Juge N (2012) Microbial adhesins to gastrointestinal mucus. Trends Microbiol 20:30–39. doi: https://doi.org/10.1016/j.tim.2011.10.001

Kamada N, Nunez G (2013) Role of the gut microbiota in the development and function of lymphoid cells. J Immunol 190(4):1389–1395

Kamada N, Chen GY, Inohara N, Núñez G (2013) Control of pathogens and pathobionts by the gut microbiota. Nat Immunol 14(7):685–690. doi: https://doi.org/10.1038/ni.2608

Kanamori Y, Ishimaru K, Nanno M et al (1996) Identification of novel lymphoid tissues in murine intestinal mucosa where clusters of c-kit + IL-7R + Thy1 + lympho-hemopoietic progenitors develop. J Exp Med 184:1449–1459

Kanwar SS, Walia S, Sharma S (2016) Impact of probiotics and gut microbiota on host behavior. In: Garg N, Abdel-Aziz SM, Aeron A (eds) Microbes in food and health. Springer, Cham, pp 29–41. doi: https://doi.org/10.1007/978-3-319-25277-3_2

Kaplan JL, Shi HN, Walker WA (2011) The role of microbes in developmental immunologic programming. Pediatr Res 69:465–472. doi: https://doi.org/10.1203/PDR.0b013e318217638a

Keenan K, Gunthorpe D, Grace D (2007) Parsing the relations between SES and stress reactivity: examining individual differences in neonatal stress response. Infant Behav Dev 30:134–145. doi: https://doi.org/10.1016/j.infbeh.2006.08.001

Keenan K, Bartlett TQ, Nijland M et al (2013) Poor nutrition during pregnancy and lactation negatively affects neurodevelopment of the offspring: evidence from a translational primate model. Am J Clin Nutr 98:396–402. doi: https://doi.org/10.3945/ajcn.112.040352

Kellogg TF (1971) Microbiological aspects of enterohepatic neutral sterol and bile acid metabolism. Fed Proc 30:1808–1814

Kieper WC, Troy A, Burghardt JT et al (2005) Recent immune status determines the source of antigens that drive homeostatic T cell expansion. J Immunol Baltim Md 1950 174:3158–3163

Kim J, Park W (2013) Indole inhibits bacterial quorum sensing signal transmission by interfering with quorum sensing regulator folding. Microbiology 159:2616–2625. doi: https://doi.org/10.1099/mic.0.070615-0

Klein RE, Arenales P, Delgado H et al (1976) Effects of maternal nutrition on fetal growth and infant development. Bull Pan Am Health Organ 10:301–306

Koboziev I, Webb CR, Furr KL, Grisham MB (2014) Role of the enteric microbiota in intestinal homeostasis and inflammation. Free Radic Biol Med 68:122–133. doi: https://doi.org/10.1016/j.freeradbiomed.2013.11.008

Koenig JE, Spor A, Scalfone N et al (2011) Succession of microbial consortia in the developing infant gut microbiome. Proc Natl Acad Sci USA 108(Suppl 1):4578–4585. doi: https://doi.org/10.1073/pnas.1000081107

Koeth RA, Wang Z, Levison BS et al (2013) Intestinal microbiota metabolism of L-carnitine, a nutrient in red meat, promotes atherosclerosis. Nat Med 19:576–585. doi: https://doi.org/10.1038/nm.3145

Kordalewska M, Markuszewski MJ (2015) Metabolomics in cardiovascular diseases. J Pharm Biomed Anal 113:121–136. doi: https://doi.org/10.1016/j.jpba.2015.04.021

Kostic AD, Gevers D, Pedamallu CS et al (2012) Genomic analysis identifies association of *Fusobacterium* with colorectal carcinoma. Genome Res 22:292–298. doi: https://doi.org/10.1101/gr.126573.111

Kumar J, Kumar M, Gupta S et al (2016) An improved methodology to overcome key issues in human fecal metagenomic DNA extraction. Genom Proteom Bioinform 14:371–378. doi: https://doi.org/10.1016/j.gpb.2016.06.002

Kumar J, Kumar M, Pandey R, Chauhan NS (2017) Physiopathology and management of gluten-induced celiac disease: celiac disease J Food Sci 82:270–277. doi: https://doi.org/10.1111/1750-3841.13612

Kumar Mondal A, Kumar J, Pandey R et al (2017) Comparative genomics of host–symbiont and free-living *Oceanobacillus* species. Genome Biol Evol 9:1175–1182. doi: https://doi.org/10.1093/gbe/evx076

Kunz C, Kuntz S, Rudloff S (2009) Intestinal flora. Adv Exp Med Biol 639:67–79. doi: https://doi.org/10.1007/978-1-4020-8749-3_6

Lajoie SF, Bank S, Miller TL, Wolin MJ (1988) Acetate production from hydrogen and [13C]carbon dioxide by the microflora of human feces. Appl Environ Microbiol 54:2723–2727

Lampe JW (2009) Is equol the key to the efficacy of soy foods? Am J Clin Nutr 89:1664S–1667S. doi: https://doi.org/10.3945/ajcn.2009.26736T

Landete JM, Arqués J, Medina M et al (2016) Bioactivation of phytoestrogens: intestinal bacteria and health. Crit Rev Food Sci Nutr 56:1826–1843. doi: https://doi.org/10.1080/10408398.2013.789823

LeBlanc JG, Milani C, de Giori GS et al (2013) Bacteria as vitamin suppliers to their host: a gut microbiota perspective. Curr Opin Biotechnol 24:160–168. doi: https://doi.org/10.1016/j.copbio.2012.08.005

Leloup J, Fossing H, Kohls K et al (2009) Sulfate-reducing bacteria in marine sediment (Aarhus Bay, Denmark): abundance and diversity related to geochemical zonation. Environ Microbiol 11:1278–1291. doi: https://doi.org/10.1111/j.1462-2920.2008.01855.x

Ley RE, Bäckhed F, Turnbaugh P et al (2005) Obesity alters gut microbial ecology. Proc Natl Acad Sci USA 102:11070–11075. doi: https://doi.org/10.1073/pnas.0504978102

Ley RE, Peterson DA, Gordon JI (2006) Ecological and evolutionary forces shaping microbial diversity in the human intestine. Cell 124:837–848. doi: https://doi.org/10.1016/j.cell.2006.02.017

Lin L, Zhang J (2017) Role of intestinal microbiota and metabolites on gut homeostasis and human diseases. BMC Immunol 18(1):2. doi: https://doi.org/10.1186/s12865-016-0187-3

Lou HC, Hansen D, Nordentoft M et al (1994) Prenatal stressors of human life affect fetal brain development. Dev Med Child Neurol 36:826–832

Loubinoux J, Valente FMA, Pereira IAC et al (2002) Reclassification of the only species of the genus *Desulfomonas, Desulfomonas pigra*, as *Desulfovibrio piger* comb. nov. Int J Syst Evol Microbiol 52:1305–1308. doi: https://doi.org/10.1099/00207713-52-4-1305

Louis P, Duncan SH, McCrae SI et al (2004) Restricted distribution of the butyrate kinase pathway among butyrate-producing bacteria from the human colon. J Bacteriol 186:2099–2106

Louis P, Hold GL, Flint HJ (2014a) The gut microbiota, bacterial metabolites and colorectal cancer. Nat Rev Microbiol 12:661–672. doi: https://doi.org/10.1038/nrmicro3344

Louis P, Hold GL, Flint HJ (2014b) The gut microbiota, bacterial metabolites and colorectal cancer. Nat Rev Microbiol 12:661–672. doi: https://doi.org/10.1038/nrmicro3344

Lowe LP, Metzger BE, Lowe WL et al (2010) Inflammatory mediators and glucose in pregnancy: results from a subset of the Hyperglycemia and Adverse Pregnancy Outcome (HAPO) Study. J Clin Endocrinol Metab 95:5427–5434. doi: https://doi.org/10.1210/jc.2010-1662

Macfarlane S, Macfarlane GT (2006) Composition and metabolic activities of bacterial biofilms colonizing food residues in the human gut. Appl Environ Microbiol 72:6204–6211. doi: https://doi.org/10.1128/AEM.00754-06

Macfarlane GT, Cummings JH, Allison C (1986) Protein degradation by human intestinal bacteria. J Gen Microbiol 132:1647–1656. doi: https://doi.org/10.1099/00221287-132-6-1647

Machiels K, Joossens M, Sabino J et al (2014) A decrease of the butyrate-producing species *Roseburia hominis* and *Faecalibacterium prausnitzii* defines dysbiosis in patients with ulcerative colitis. Gut 63:1275–1283. doi: https://doi.org/10.1136/gutjnl-2013-304833

Magnúsdóttir S, Ravcheev D, de Crécy-Lagard V, Thiele I (2015) Systematic genome assessment of B-vitamin biosynthesis suggests co-operation among gut microbes. Front Genet 6:148. doi: https://doi.org/10.3389/fgene.2015.00148

Manach C, Scalbert A, Morand C et al (2004) Polyphenols: food sources and bioavailability. Am J Clin Nutr 79:727–747

Marín L, Miguélez EM, Villar CJ, Lombó F (2015) Bioavailability of dietary polyphenols and gut microbiota metabolism: antimicrobial properties. BioMed Res Int 2015:905215. doi: https://doi.org/10.1155/2015/905215

Marteau P, Pochart P, Doré J et al (2001) Comparative study of bacterial groups within the human cecal and fecal microbiota. Appl Environ Microbiol 67:4939–4942

Monika MKV, Ahmed V, Chauhan NS (2017) Human gut microbiome: an imperative element for human survival. Curr Trends Biomedical Eng Biosci 6(1):555680. doi: https://doi.org/10.19080/CTBEB.2017.06.555680

Moroni F (1999) Tryptophan metabolism and brain function: focus on kynurenine and other indole metabolites. Eur J Pharmacol 375:87–100

Mortensen PB, Clausen MR (1996) Short-chain fatty acids in the human colon: relation to gastrointestinal health and disease. Scand J Gastroenterol Suppl 216:132–148

Muir JG, Yeow EGW, Keogh J et al (2004) Combining wheat bran with resistant starch has more beneficial effects on fecal indexes than does wheat bran alone. Am J Clin Nutr 79:1020–1028

Murri M, Leiva I, Gomez-Zumaquero JM et al (2013) Gut microbiota in children with type 1 diabetes differs from that in healthy children: a case–control study. BMC Med 11:46. doi: https://doi.org/10.1186/1741-7015-11-46

Muyzer G, Stams AJM (2008) The ecology and biotechnology of sulphate-reducing bacteria. Nat Rev Microbiol 6:441–454. doi: https://doi.org/10.1038/nrmicro1892

Neish AS (2009) Microbes in gastrointestinal health and disease. Gastroenterology 136:65–80. doi: https://doi.org/10.1053/j.gastro.2008.10.080

Norris V, Molina F, Gewirtz AT (2013) Hypothesis: bacteria control host appetites. J Bacteriol 195(3):411–416

O'Hara AM, Shanahan F (2006) The gut flora as a forgotten organ. EMBO Rep 7:688–693. doi: https://doi.org/10.1038/sj.embor.7400731

Ouwehand AC, Derrien M, de Vos W et al (2005) Prebiotics and other microbial substrates for gut functionality. Curr Opin Biotechnol 16:212–217. doi: https://doi.org/10.1016/j.copbio.2005.01.007

Peek RM, Blaser MJ (2002) *Helicobacter pylori* and gastrointestinal tract adenocarcinomas. Nat Rev Cancer 2:28–37. doi: https://doi.org/10.1038/nrc703

Penders J, Thijs C, Vink C et al (2006) Factors influencing the composition of the intestinal microbiota in early infancy. Pediatrics 118:511–521. doi: https://doi.org/10.1542/peds.2005-2824

Pérez-Jiménez J, Fezeu L, Touvier M et al (2011) Dietary intake of 337 polyphenols in French adults. Am J Clin Nutr 93:1220–1228. doi: https://doi.org/10.3945/ajcn.110.007096

Perry GH, Dominy NJ, Claw KG et al (2007) Diet and the evolution of human amylase gene copy number variation. Nat Genet 39:1256–1260. doi: https://doi.org/10.1038/ng2123

Pickard JM, Maurice CF, Kinnebrew MA et al (2014) Rapid fucosylation of intestinal epithelium sustains host-commensal symbiosis in sickness. Nature 514:638–641. doi: https://doi.org/10.1038/nature13823

Praveen P, Jordan F, Priami C, Morine MJ (2015) The role of breastfeeding in infant immune system: a systems perspective on the intestinal microbiome. Microbiome. doi: https://doi.org/10.1186/s40168-015-0104-7

Prentiss PG, Rosen H, Brown N et al (1961) The metabolism of choline by the germfree rat. Arch Biochem Biophys 94:424–429

Principi N, Esposito S (2016) Gut microbiota and central nervous system development. J Infect 73(6):536–546

Pryde SE, Duncan SH, Hold GL et al (2002) The microbiology of butyrate formation in the human colon. FEMS Microbiol Lett 217:133–139

Qin J, Li R, Raes J et al (2010) A human gut microbial gene catalogue established by metagenomic sequencing. Nature 464:59–65. doi: https://doi.org/10.1038/nature08821

Quartieri A, García-Villalba R, Amaretti A et al (2016) Detection of novel metabolites of flaxseed lignans in vitro and in vivo. Mol Nutr Food Res 60:1590–1601. doi: https://doi.org/10.1002/mnfr.201500773

Rechner AR, Smith MA, Kuhnle G et al (2004) Colonic metabolism of dietary polyphenols: influence of structure on microbial fermentation products. Free Radic Biol Med 36:212–225

Reichardt N, Duncan SH, Young P et al (2014) Phylogenetic distribution of three pathways for propionate production within the human gut microbiota. ISME J 8:1323–1335. doi: https://doi.org/10.1038/ismej.2014.14

Reinhardt C, Bergentall M, Greiner TU et al (2012) Tissue factor and PAR1 promote microbiota-induced intestinal vascular remodelling. Nature 483:627–631. doi: https://doi.org/10.1038/nature10893

Ridlon JM, Kang D-J, Hylemon PB (2006) Bile salt biotransformations by human intestinal bacteria. J Lipid Res 47:241–259. doi: https://doi.org/10.1194/jlr.R500013-JLR200

Romano KA, Vivas EI, Amador-Noguez D, Rey FE (2015) Intestinal microbiota composition modulates choline bioavailability from diet and accumulation of the proatherogenic metabolite trimethylamine-*N*-oxide. mBio 6:e02481. doi: https://doi.org/10.1128/mBio.02481-14

Rosenberg E, Zilber-Rosenberg I (2011) Symbiosis and development: the hologenome concept. Birth Defects Res Part C Embryo Today Rev 93:56–66. doi: https://doi.org/10.1002/bdrc.20196

Rossi M, Corradini C, Amaretti A et al (2005) Fermentation of fructooligosaccharides and inulin by bifidobacteria: a comparative study of pure and fecal cultures. Appl Environ Microbiol 71:6150–6158. doi: https://doi.org/10.1128/AEM.71.10.6150-6158.2005

Ruma M, Horton A, Boggess K et al (2007) 221: Systemic inflammation: Its role in periodontal disease and preterm birth. Am J Obstet Gynecol 197:S73. doi: https://doi.org/10.1016/j.ajog.2007.10.234

Russell WR, Duncan SH, Scobbie L et al (2013) Major phenylpropanoid-derived metabolites in the human gut can arise from microbial fermentation of protein. Mol Nutr Food Res 57:523–535. doi: https://doi.org/10.1002/mnfr.201200594

Said HM (2013) Recent advances in transport of water-soluble vitamins in organs of the digestive system: a focus on the colon and the pancreas. Am J Physiol Gastrointest Liver Physiol 305:G601–G610. doi: https://doi.org/10.1152/ajpgi.00231.2013

Sakata H, Yoshioka H, Fujita K (1985) Development of the intestinal flora in very low birth weight infants compared to normal full-term newborns. Eur J Pediatr 144:186–190

Samuel BS, Gordon JI (2006) A humanized gnotobiotic mouse model of host–archaeal–bacterial mutualism. Proc Natl Acad Sci USA 103:10011–10016. doi: https://doi.org/10.1073/pnas.0602187103

Sandyk R (1992) L-tryptophan in neuropsychiatric disorders: a review. Int J Neurosci 67:127–144

Sapolsky RM, Romero LM, Munck AU (2000) How do glucocorticoids influence stress responses? Integrating permissive, suppressive, stimulatory, and preparative actions. Endocr Rev 21:55–89. doi: https://doi.org/10.1210/edrv.21.1.0389

Savage DC (1970) Associations of indigenous microorganisms with gastrointestinal mucosal epithelia. Am J Clin Nutr 23(11):1495–1501

Scanlan PD, Shanahan F, Marchesi JR (2009) Culture-independent analysis of *desulfovibrios* in the human distal colon of healthy, colorectal cancer and polypectomized individuals. FEMS Microbiol Ecol 69:213–221. doi: https://doi.org/10.1111/j.1574-6941.2009.00709.x

Schloss PD, Handelsman J (2004) Status of the microbial census. Microbiol Mol Biol Rev MMBR 68:686–691. doi: https://doi.org/10.1128/MMBR.68.4.686-691.2004

Schneider ML, Moore CF, Kraemer GW et al (2002) The impact of prenatal stress, fetal alcohol exposure, or both on development: perspectives from a primate model. Psychoneuroendocrinology 27:285–298

Scott KP, Martin JC, Campbell G et al (2006) Whole-genome transcription profiling reveals genes up-regulated by growth on fucose in the human gut bacterium "*Roseburia inulinivorans*".J Bacteriol 188:4340–4349. doi: https://doi.org/10.1128/JB.00137-06

Scott KP, Duncan SH, Flint HJ (2008) Dietary fibre and the gut microbiota. Nutr Bull 33:201–211. doi: https://doi.org/10.1111/j.1467-310.2008.00706.x

Seksik P, Rigottier-Gois L, Gramet G et al (2003) Alterations of the dominant faecal bacterial groups in patients with Crohn's disease of the colon. Gut 52:237–242

Shanahan F (2013) The colonic microbiota in health and disease. Curr Opin Gastroenterol 29:49–54. doi: https://doi.org/10.1097/MOG.0b013e32835a3493

Sharma R, Young C, Neu J (2010) Molecular modulation of intestinal epithelial barrier: contribution of microbiota. J Biomed Biotechnol 2010:305879. doi: https://doi.org/10.1155/2010/305879.

Sjögren K, Engdahl C, Henning P et al (2012) The gut microbiota regulates bone mass in mice. J Bone Miner Res Off J Am Soc Bone Miner Res 27:1357–1367. doi: https://doi.org/10.1002/jbmr.1588

Smith EA, Macfarlane GT (1996) Enumeration of human colonic bacteria producing phenolic and indolic compounds: effects of pH, carbohydrate availability and retention time on dissimilatory aromatic amino acid metabolism. J Appl Bacteriol 81:288–302

Spencer MD, Hamp TJ, Reid RW et al (2011) Association between composition of the human gastrointestinal microbiome and development of fatty liver with choline deficiency. Gastroenterology 140:976–986. doi: https://doi.org/10.1053/j.gastro.2010.11.049

Stappenbeck TS, Hooper LV, Gordon JI (2002) Developmental regulation of intestinal angiogenesis by indigenous microbes via Paneth cells. Proc Natl Acad Sci USA 99:15451–15455. doi: https://doi.org/10.1073/pnas.202604299

Suarez F, Furne J, Springfield J, Levitt M (1997) Insights into human colonic physiology obtained from the study of flatus composition. Am J Physiol 272:G1028–G1033

Tana C, Umesaki Y, Imaoka A et al (2010) Altered profiles of intestinal microbiota and organic acids may be the origin of symptoms in irritable bowel syndrome. Neurogastroenterol Motil Off J Eur Gastrointest Motil Soc 22:512–519, e114–115. doi: https://doi.org/10.1111/j.1365-2982.2009.01427.x

Thadepalli H, Lou MA, Bach VT et al (1979) Microflora of the human small intestine. Am J Surg 138:845–850

Tomas-Barberan F, García-Villalba R, Quartieri A et al (2014) In vitro transformation of chlorogenic acid by human gut microbiota. Mol Nutr Food Res 58:1122–1131. doi: https://doi.org/10.1002/mnfr.201300441

Tomás-Barberán FA, González-Sarrías A, García-Villalba R et al (2017) Urolithins, the rescue of "old" metabolites to understand a "new" concept: Metabotypes as a nexus among phenolic metabolism, microbiota dysbiosis, and host health status. Mol Nutr Food Res. doi: https://doi.org/10.1002/mnfr.201500901

Turnbaugh PJ, Ley RE, Mahowald MA et al (2006) An obesity-associated gut microbiome with increased capacity for energy harvest. Nature 444:1027–1031. doi: https://doi.org/10.1038/nature05414

Turroni F, Ribbera A, Foroni E et al (2008) Human gut microbiota and bifidobacteria: from composition to functionality. Antonie Van Leeuwenhoek 94:35–50. doi: https://doi.org/10.1007/s10482-008-9232-4

Ubeda C, Djukovic A, Isaac S (2017) Roles of the intestinal microbiota in pathogen protection. Clin Transl Immunol 6(2):e128. doi: https://doi.org/10.1038/cti.2017.2.

Uematsu S, Fujimoto K, Jang MH et al (2008) Regulation of humoral and cellular gut immunity by lamina propria dendritic cells expressing toll-like receptor 5. Nat Immunol 9:769–776. doi: https://doi.org/10.1038/ni.1622

Vaishnava S, Yamamoto M, Severson KM et al (2011) The antibacterial lectin RegIIIgamma promotes the spatial segregation of microbiota and host in the intestine. Science 334:255–258. doi: https://doi.org/10.1126/science.1209791

Vital M, Howe AC, Tiedje JM (2014) Revealing the bacterial butyrate synthesis pathways by analyzing (meta)genomic data. mBio 5:e00889. doi: https://doi.org/10.1128/mBio. 00889–14

Wadhwa PD, Sandman CA, Porto M et al (1993) The association between prenatal stress and infant birth weight and gestational age at birth: a prospective investigation. Am J Obstet Gynecol 169:858–865

Wall R, Ross RP, Ryan CA et al (2009) Role of gut microbiota in early infant development. Clin Med Pediatr 3:45–54

Wang X, Heazlewood SP, Krause DO, Florin THJ (2003) Molecular characterization of the microbial species that colonize human ileal and colonic mucosa by using 16S rDNA sequence analysis. J Appl Microbiol 95:508–520. doi: https://doi.org/10.1046/j.1365-2672.2003.02005.x

Wang J, Zhao H, Kong W et al (2010) Microcalorimetric assay on the antimicrobial property of five hydroxyanthraquinone derivatives in rhubarb (*Rheum palmatum* L.) to Bifidobacterium adolescentis. Phytomed Int J Phytother Phytopharm 17:684–689. doi: https://doi.org/10.1016/j.phymed.2009.10.009

Wang Z, Klipfell E, Bennett BJ et al (2011) Gut flora metabolism of phosphatidylcholine promotes cardiovascular disease. Nature 472:57–63. doi: https://doi.org/10.1038/nature09922

Wauson EM, Lorente-Rodríguez A, Cobb MH (2013) Minireview: nutrient sensing by G protein-coupled receptors. Mol Endocrinol Baltim Md 27:1188–1197. doi: https://doi.org/10.1210/me.2013-1100

Weinstock M (1997) Does prenatal stress impair coping and regulation of hypothalamic–pituitary–adrenal axis? Neurosci Biobehav Rev 21:1–10

Weinstock M (2005) The potential influence of maternal stress hormones on development and mental health of the offspring. Brain Behav Immun 19:296–308. doi: https://doi.org/10.1016/j.bbi.2004.09.006

Weir TL, Manter DK, Sheflin AM et al (2013) Stool microbiome and metabolome differences between colorectal cancer patients and healthy adults. PloS One 8:e70803. doi: https://doi.org/10.1371/journal.pone.0070803

Wikoff WR, Anfora AT, Liu J et al (2009) Metabolomics analysis reveals large effects of gut microflora on mammalian blood metabolites. Proc Natl Acad Sci USA 106:3698–3703. doi: https://doi.org/10.1073/pnas.0812874106

Wolf PG, Biswas A, Morales SE et al (2016) H2 metabolism is widespread and diverse among human colonic microbes. Gut Microbes 7:235–245. doi: https://doi.org/10.1080/19490976.2016.1182288

Woollett LA, Buckley DD, Yao L et al (2003) Effect of ursodeoxycholic acid on cholesterol absorption and metabolism in humans. J Lipid Res 44:935–942. doi: https://doi.org/10.1194/jlr.M200478-JLR200

Xu J, Gordon JI (2003) Honor thy symbionts. Proc Natl Acad Sci USA 100:10452–10459. doi: https://doi.org/10.1073/pnas.1734063100

Yang EV, Glaser R (2002) Stress-induced immunomodulation and the implications for health. Int Immunopharmacol 2:315–324

Yeoh N, Burton JP, Suppiah P et al (2013) The role of the microbiome in rheumatic diseases. Curr Rheumatol Rep 15:314. doi: https://doi.org/10.1007/s11926-012-0314-y

Yu LC, Wang JT, Wei SC, Ni YH (2012) Host-microbial interactions and regulation of intestinal epithelial barrier function: from physiology to pathology. World J Gastrointest Pathophysiol 3(1):27

Zelante T, Iannitti RG, Cunha C et al (2013) Tryptophan catabolites from microbiota engage aryl hydrocarbon receptor and balance mucosal reactivity via interleukin-22. Immunity 39:372–385. doi: https://doi.org/10.1016/j.immuni.2013.08.003

Zhang Y-J, Li S, Gan R-Y et al (2015) Impacts of gut bacteria on human health and diseases. Int J Mol Sci 16:7493–7519. doi: https://doi.org/10.3390/ijms16047493

Zijlmans MAC, Korpela K, Riksen-Walraven JM et al (2015) Maternal prenatal stress is associated with the infant intestinal microbiota. Psychoneuroendocrinology 53:233–245. doi: https://doi.org/10.1016/j.psyneuen.2015.01.006